Name of Student: Minisha Singh	**PRN: 25070122292**
Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 24	
TITLE : JDBC with GUI	

Problem Statement

Develop a Java GUI application integrated with JDBC to perform database operations."

Learning Objectives

To integrate Java GUI applications with databases using JDBC.

Exercise

1. **Create a Library Management System GUI connected with database**

Code:

```
import javax.swing.*;
import java.awt.*;
import java.sql.*;

public class LibraryManagement extends JFrame {

    JTextField idField, nameField, authorField;
    JTextArea output;

    String url = "jdbc:mysql://localhost:3306/college";
    String user = "root";
    String password = "mini@12";

    LibraryManagement() {

        setTitle("Library Management System");
        setSize(500, 500);
        setLayout(new FlowLayout());

        add(new JLabel("Book ID:"));
```

```
    idField = new JTextField(20);
    add(idField);

    add(new JLabel("Book Name:"));
    nameField = new JTextField(20);
    add(nameField);

    add(new JLabel("Author:"));
    authorField = new JTextField(20);
    add(authorField);

    JButton addButton = new JButton("Add Book");
    JButton viewButton = new JButton("View Books");
    JButton deleteButton = new JButton("Delete Book");

    add(addButton);
    add(viewButton);
    add(deleteButton);

    output = new JTextArea(15, 40);
    add(new JScrollPane(output));

    addButton.addActionListener(e -> addBook());
    viewButton.addActionListener(e -> viewBooks());
    deleteButton.addActionListener(e -> deleteBook());

    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setVisible(true);
}

void addBook() {

    try {
        Connection con = DriverManager.getConnection(url, user, password);

        String query = "INSERT INTO library VALUES (?, ?, ?)";
        PreparedStatement ps = con.prepareStatement(query);

        ps.setInt(1, Integer.parseInt(idField.getText()));
        ps.setString(2, nameField.getText());
        ps.setString(3, authorField.getText());

        ps.executeUpdate();
```

```
            output.setText("Book added successfully.");

            con.close();

        } catch (Exception e) {
            output.setText(e.getMessage());
        }
    }

    void viewBooks() {

        try {
            Connection con = DriverManager.getConnection(url, user, password);

            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM library");

            output.setText("Book Records:\n\n");

            while (rs.next()) {
                output.append(
                    "ID: " + rs.getInt("book_id") +
                    " | Name: " + rs.getString("book_name") +
                    " | Author: " + rs.getString("author") + "\n"
                );
            }

            con.close();

        } catch (Exception e) {
            output.setText(e.getMessage());
        }
    }

    void deleteBook() {

        try {
            Connection con = DriverManager.getConnection(url, user, password);

            String query = "DELETE FROM library WHERE book_id = ?";
            PreparedStatement ps = con.prepareStatement(query);

            ps.setInt(1, Integer.parseInt(idField.getText()));
```

```
            ps.executeUpdate();

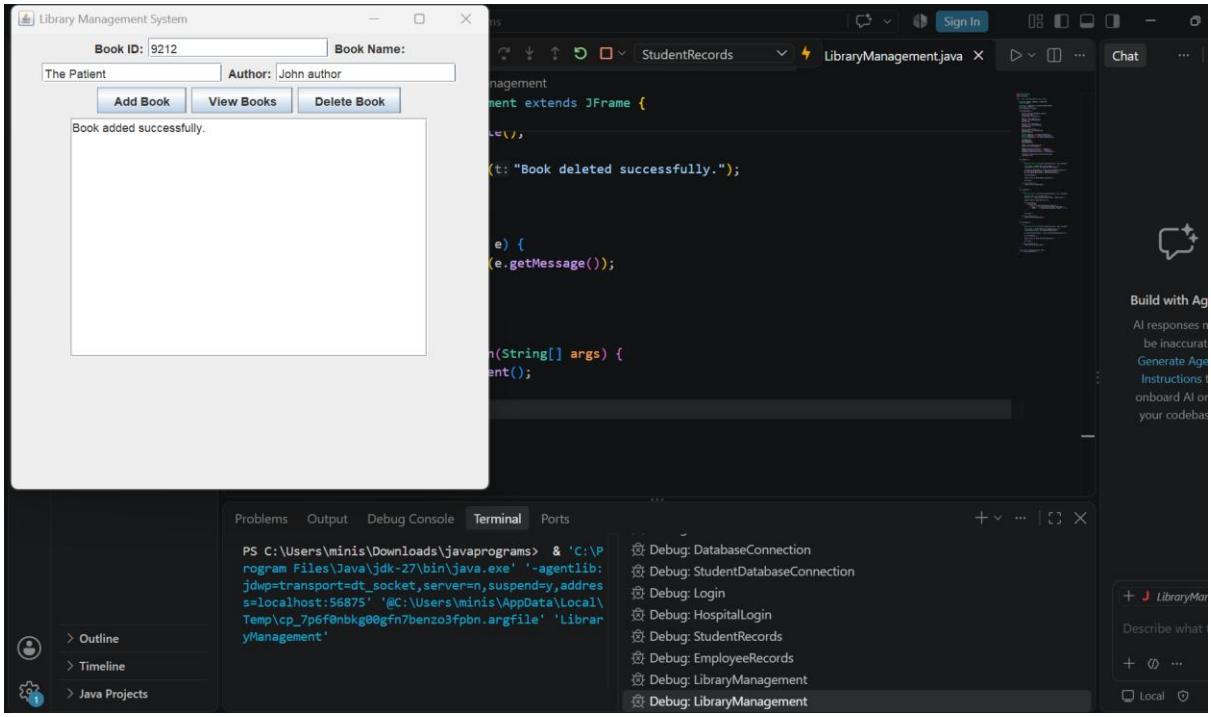
            output.setText("Book deleted successfully.");

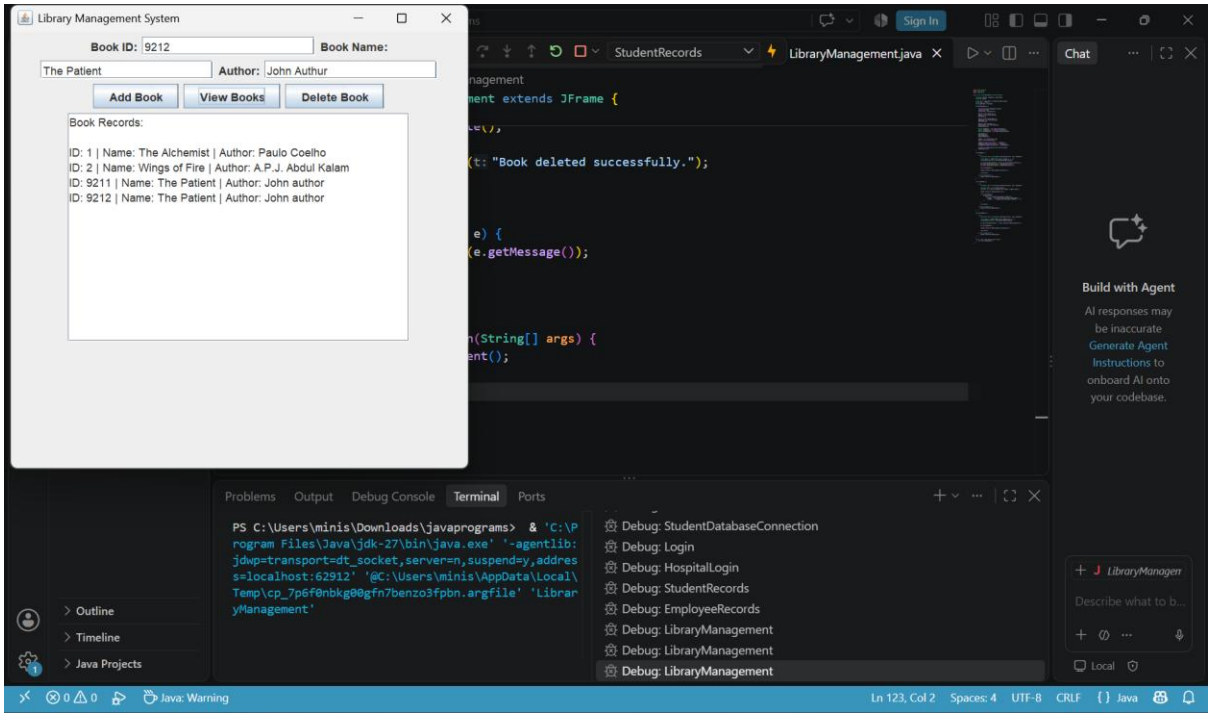
            con.close();

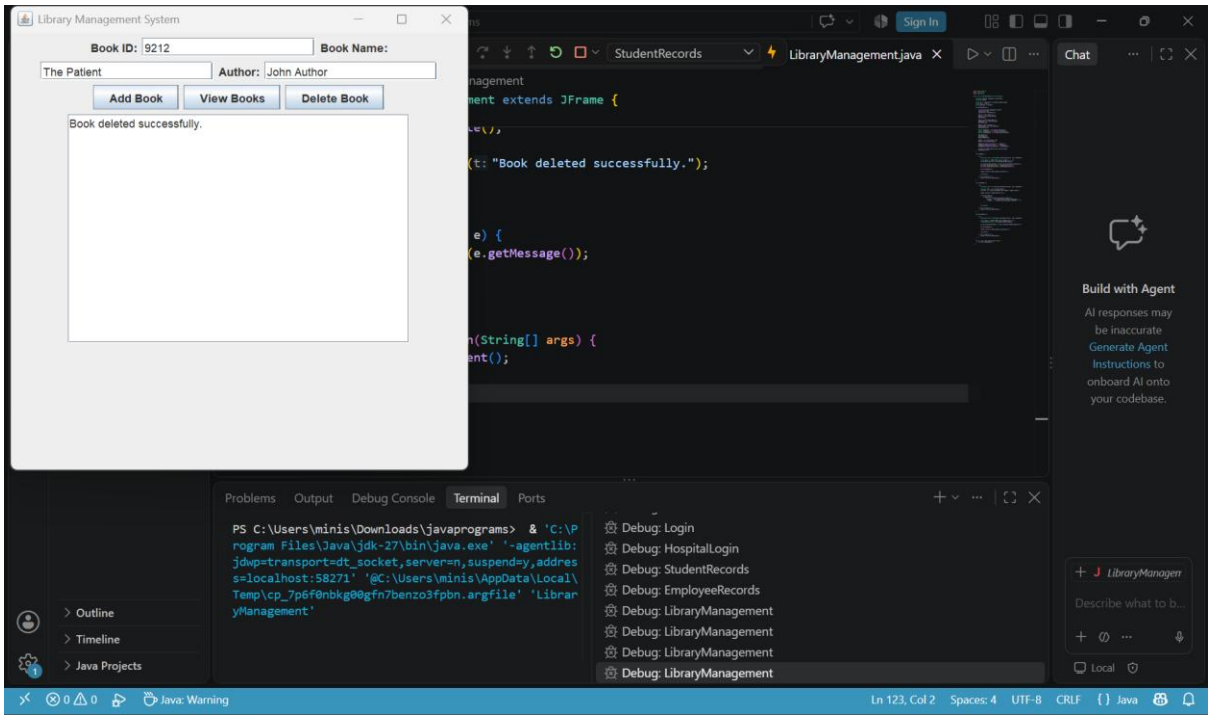
        } catch (Exception e) {
            output.setText(e.getMessage());
        }
    }

    public static void main(String[] args) {
        new LibraryManagement();
    }
}
```

Output:




Library Management System
Book ID: 9212
Book Name:
The Patient
Author: John Authur
Add Book
View Books
Delete Book
Book Records:
ID: 1 | Name: The Alchemist | Author: Paulo Coelho
ID: 2 | Name: Wings of Fire | Author: A.P.J. Abdul Kalam
ID: 9211 | Name: The Patient | Author: John author
ID: 9212 | Name: The Patient | Author: John author
StudentRecords
LibraryManagement.java
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\minis\Downloads\javaprograms> & 'C:\Program Files\Java\jdk-27\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:62912' '@C:\Users\minis\AppData\Local\Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'LibraryManagement'
Debug: StudentDatabaseConnection
Debug: Login
Debug: HospitalLogin
Debug: StudentRecords
Debug: EmployeeRecords
Debug: LibraryManagement
Debug: LibraryManagement
Debug: LibraryManagement
Outline
Timeline
Java Projects
Java: Warning
Ln 123, Col 2
Spaces: 4
UTF-8
CRLF
Java


Library Management System
Book ID: 9212
Book Name:
The Patient
Author: John Author
Add Book
View Books
Delete Book
Book deleted successfully.
StudentRecords
LibraryManagement.java
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\minis\Downloads\javaprograms> & 'C:\Program Files\Java\jdk-27\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:58271' '@C:\Users\minis\AppData\Local\Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'LibraryManagement'
Debug: Login
Debug: HospitalLogin
Debug: StudentRecords
Debug: EmployeeRecords
Debug: LibraryManagement
Debug: LibraryManagement
Debug: LibraryManagement
Debug: LibraryManagement
Outline
Timeline
Java Projects
Java: Warning
Ln 123, Col 2
Spaces: 4
UTF-8
CRLF
Java

2. **Create a Book Issue Tracking System GUI application using Java Swing connected to a database. The system should store book ID, student name, issue date, and return date, and allow record management.**

Code:

```
import javax.swing.*;
import java.awt.*;
import java.sql.*;

public class BookIssueTracking extends JFrame {

   JTextField bookId, studentName, issueDate, returnDate;
   JTextArea output;

   String url = "jdbc:mysql://localhost:3306/college";
   String user = "root";
   String password = "mini@12";

   BookIssueTracking() {
      setTitle("Book Issue Tracking System");
      setSize(550, 500);
      setLayout(new FlowLayout());

      add(new JLabel("Book ID:"));
      bookId = new JTextField(20);
      add(bookId);

      add(new JLabel("Student Name:"));
      studentName = new JTextField(20);
      add(studentName);

      add(new JLabel("Issue Date (YYYY-MM-DD):"));
      issueDate = new JTextField(20);
      add(issueDate);

      add(new JLabel("Return Date (YYYY-MM-DD):"));
      returnDate = new JTextField(20);
      add(returnDate);

      JButton addButton = new JButton("Add Record");
      JButton viewButton = new JButton("View Records");
```

```
    JButton deleteButton = new JButton("Delete Record");

    add(addButton);
    add(viewButton);
    add(deleteButton);

    output = new JTextArea(15, 45);
    add(new JScrollPane(output));

    addButton.addActionListener(e -> addRecord());
    viewButton.addActionListener(e -> viewRecords());
    deleteButton.addActionListener(e -> deleteRecord());

    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setVisible(true);
}

void addRecord() {
    try {
        Connection con = DriverManager.getConnection(url, user, password);

        String query = "INSERT INTO book_issue VALUES (?, ?, ?, ?)";
        PreparedStatement ps = con.prepareStatement(query);

        ps.setInt(1, Integer.parseInt(bookId.getText()));
        ps.setString(2, studentName.getText());
        ps.setDate(3, Date.valueOf(issueDate.getText()));
        ps.setDate(4, Date.valueOf(returnDate.getText()));

        ps.executeUpdate();

        output.setText("Book issue record added successfully.");
        con.close();

    } catch (Exception e) {
        output.setText(e.getMessage());
    }
}

void viewRecords() {
    try {
        Connection con = DriverManager.getConnection(url, user, password);

        Statement stmt = con.createStatement();
```

```
            ResultSet rs = stmt.executeQuery("SELECT * FROM book_issue");

            output.setText("Book Issue Records:\n\n");

            while (rs.next()) {
                output.append(
                    "Book ID: " + rs.getInt("book_id") +
                    " | Student: " + rs.getString("student_name") +
                    " | Issue Date: " + rs.getDate("issue_date") +
                    " | Return Date: " + rs.getDate("return_date") + "\n"
                );
            }

            con.close();

        } catch (Exception e) {
            output.setText(e.getMessage());
        }
    }

    void deleteRecord() {
        try {
            Connection con = DriverManager.getConnection(url, user, password);

            String query = "DELETE FROM book_issue WHERE book_id = ?";
            PreparedStatement ps = con.prepareStatement(query);

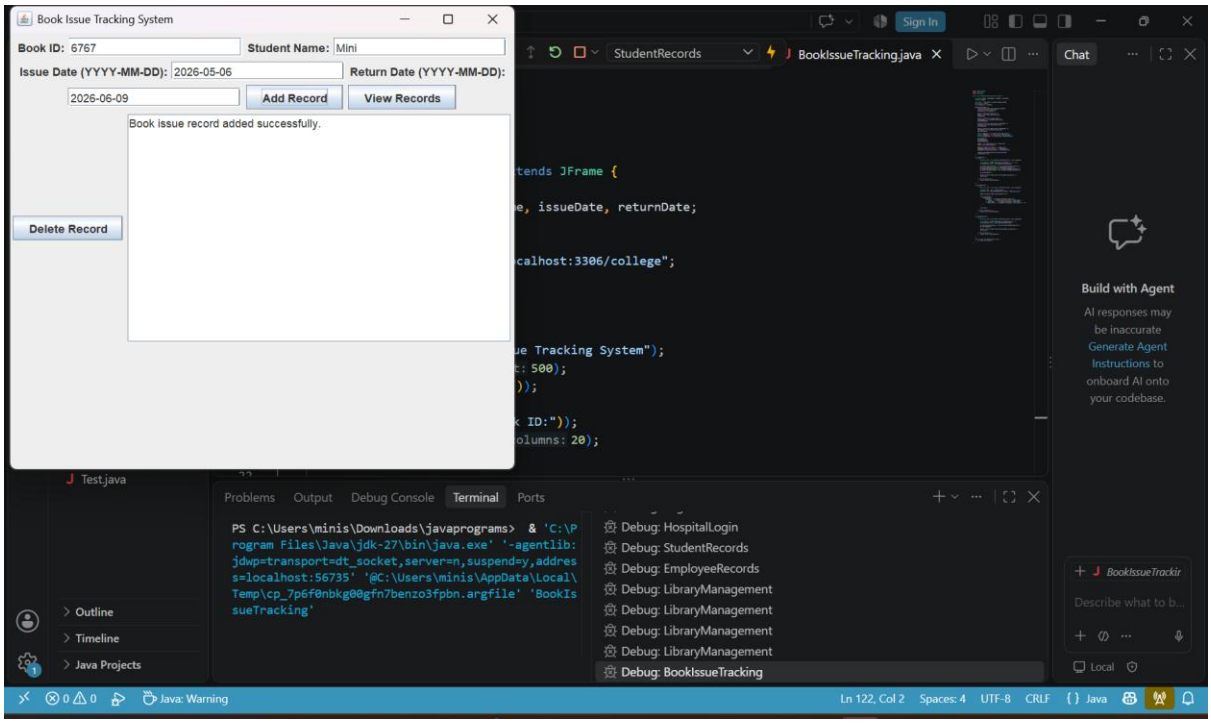
            ps.setInt(1, Integer.parseInt(bookId.getText()));
            ps.executeUpdate();

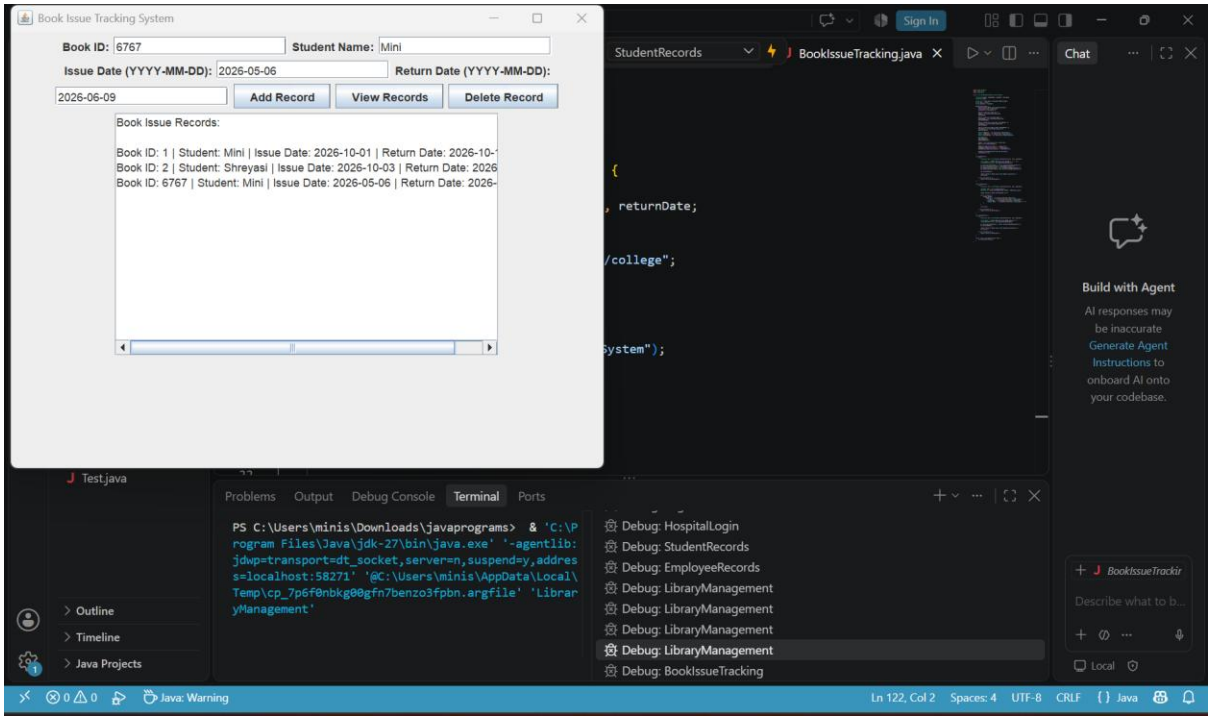
            output.setText("Book issue record deleted successfully.");
            con.close();

        } catch (Exception e) {
            output.setText(e.getMessage());
        }
    }

    public static void main(String[] args) {
        new BookIssueTracking();
    }
}
```

Output:





Conclusion

In this practical, we developed Java Swing GUI applications connected to MySQL using JDBC. We performed basic database operations such as adding, viewing, and deleting records. This helped us understand the integration of GUI components with database operations.